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AHA SCIENTIFIC STATEMENT

Clonal Hematopoiesis and Its Cardiovascular Implications: A Scientific Statement From the American Heart Association

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ABSTRACT: Clonal hematopoiesis (CH), the benign clonal expansion of hematopoietic stem cells, is often caused by somatic sequence variations in genes associated with hematologic malignancies. Over the past decade, CH has emerged as a risk factor for a wide range of cardiovascular diseases (CVDs), including atherosclerosis, heart failure, atrial fibrillation, and thrombosis. The cardiovascular risk associated with CH is heterogeneous; it varies on the basis of specific genes and variants, clone size, and various extrinsic features. Mechanistic studies suggest that CH contributes to CVDs through both gene-specific pathways and broader inflammatory processes. These include aberrant cytokine production, inflammasome activation, and other proinflammatory mechanisms, which can accelerate atherosclerosis, promote thrombogenesis, and impair vascular or myocardial function. These findings underscore the importance of addressing CH as a potential contributor to CVDs. CH is predominantly considered an age-related phenomenon, but lifelong influences on the fitness of genetic variants, including germline predispositions, obesity, chronic inflammation, and exposure to environmental toxins (eg, tobacco, certain cancer treatments), influence CH. A greater understanding of CH risk factors is therefore important for both individual and population-level risk assessments. Incorporating CH-associated risk into existing CVD risk prediction models may inform new personalized preventive or therapeutic approaches. No CH-specific therapies have proven efficacy in CVD treatment or prevention, but multiple molecular-based therapeutic hypotheses are beginning to be tested.

Key Words: AHA Scientific Statements ■ aging ■ cardiovascular diseases ■ clonal hematopoiesis ■ heart failure ■ hematopoietic stem cells ■ mutation

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Chronologic age is the dominant, unmodifiable risk factor for cardiovascular disease (CVD).[1] The incidence of CVD increases sharply with age,[2] yet the precise mechanisms by which advanced age modulates cardiovascular risk remain elusive. Aging is associated with a host of physiologic and molecular changes, including chronic low-grade inflammation (often termed "inflammaging"), oxidative stress, cellular senescence, and diminished regenerative capacity of vascular and cardiac tissues.[3–5] Recent advances point to clonal hematopoiesis (CH), itself an age-related phenomenon, as a potential missing link, offering new insight into how aging drives the heightened cardiovascular risk observed in older adults.[6,7]

CH refers to the disproportionate yet benign expansion of hematopoietic stem cell (HSC) clones, often driven by sequence variations in drivers of hematologic malignancy.[8] In 2014, 3 independent studies using whole exome sequencing of peripheral blood cells from adults without hematologic malignancies reported the increasing prevalence of CH with advancing age.[6,9,10] In these earlier studies, ≈10% of individuals >70 years of age were found to have CH with a variant allele frequency (VAF) of ≥2%, a threshold commonly used to define CH of indeterminate potential (CHIP). Later reports using more sensitive error-corrected sequencing modalities demonstrated that CH harboring leukemogenic sequence variations is nearly

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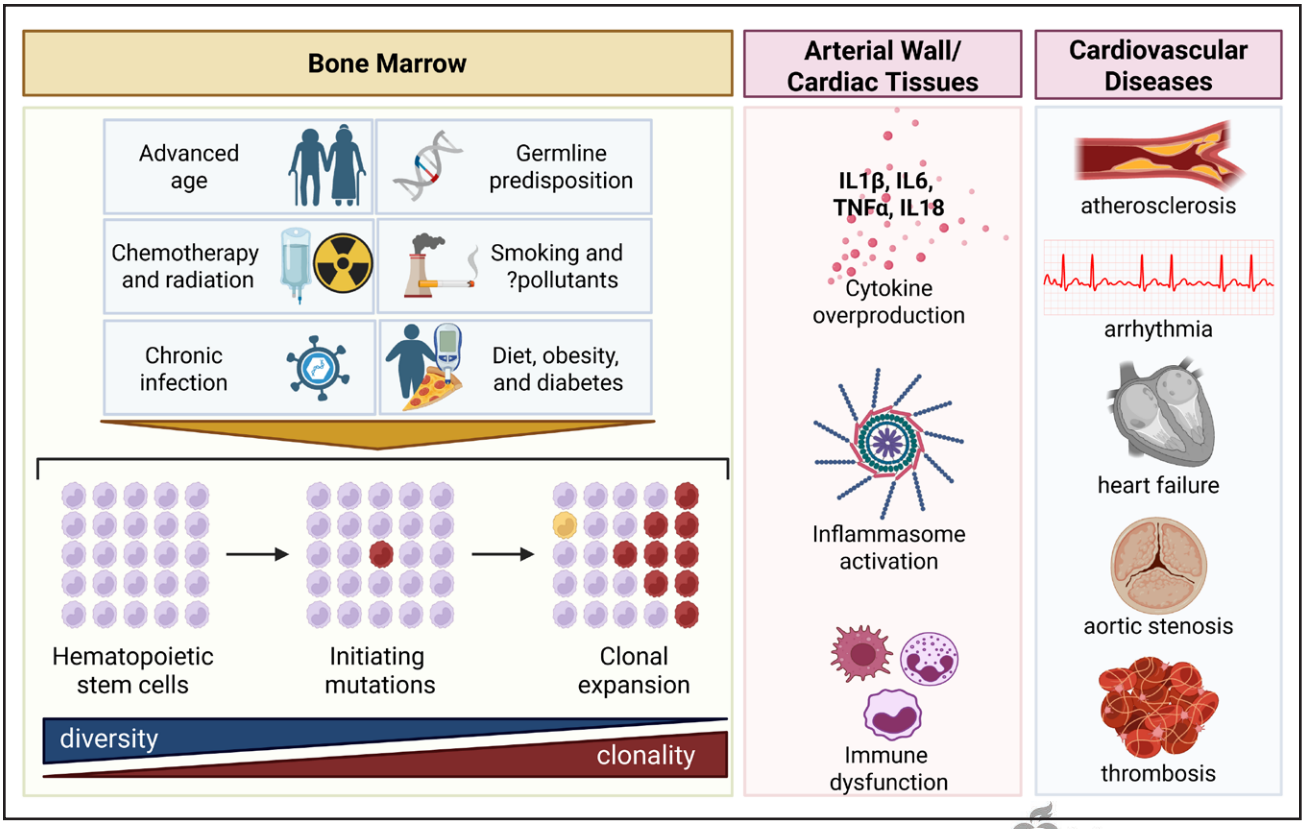

Figure 1. The association between clonal hematopoiesis and cardiovascular disease. Somatic sequence variations in hematopoietic stem cells in the bone marrow can give rise to clones over time. Various factors contribute to clonal hematopoiesis, either through mutagenesis or by promoting clonal expansion. Once these altered cells enter circulation, they can drive cytokine overproduction, inflammasome activation, and immune dysfunction in vascular and myocardial tissues, collectively increasing the risk of a broad spectrum of cardiovascular diseases. Figure created with BioRender.com.

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ubiquitous among adults >70 years of age.[11] Whereas the presence of these sequence variations increases the risk of developing hematologic malignancies, most carriers do not develop cancer. Despite the low absolute risk of malignancies, CH is associated with a 40% to 50% increase in all-cause mortality.[6,9] Several lines of evidence suggest that excess mortality in CH carriers is attributable to an increased risk of CVD, specifically incident coronary heart disease and ischemic stroke.[6] In line with this, recent studies have identified distinct epigenetic signatures associated with CH that reflect accelerated biologic aging, potentially contributing to increased CVD risk.[12,13]

With emerging evidence, there has been burgeoning interest in the relationship between CH and CVD. This scientific statement addresses the evolving definition and pathogenesis of CH, key risk factors contributing to CH pathogenesis, cardiovascular risks associated with CH, current evidence regarding gene-specific mechanisms underlying CH-related CVD risk, and the clinical implications of CH and potential therapeutic strategies (Figure 1).

EVOLVING DEFINITION OF CH

CH is defined by the presence of an expanded clonal hematopoietic stem and progenitor cell (HSPC) population in an individual without known hematologic cancer. Whereas CH is typically attributed to clonal expansion of HSCs, recent studies suggest it may also originate from downstream progenitors with aberrant self-renewal, reflecting ongoing uncertainty regarding the cellular origins of CH.[14,15] Like other acquired genetic changes that drive cancer, CH is caused by diverse genetic alterations, including single nucleotide variants,[6,9,10,16] small insertions or deletions,[6,9,10] and large copy number events.[17] Large copy number events, which sometimes may accompany other driver events, are sometimes collectively termed mosaic chromosomal alterations. Reflecting a shared origin in HSPCs, the genomic events that most commonly drive CH overlap with those that drive hematologic cancers, including both lymphoid and myeloid malignancies.[18] The genes and genetic regions that drive CH have distinct consequences on HSPC biology and resultant clinical outcomes.[19,20] Clonal expansions indicative of CH may also be detected where a clear driver variant event cannot be identified.[21] These events may reflect unrecognized drivers or genetic drift and share varied associations with traditional CH risk factors and health outcomes. In the oldest elderly individuals (eg, age >80 years), CH may represent an adaptive response to hematopoietic stress, potentially compensating for defective hematopoiesis by

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expanding clones that maintain blood cell production. However, this compensatory role is not well-established and requires further research.[22]

Various terms have evolved over the past decade to describe both the causes and consequences of CH. CHIP is defined as the presence of CH caused by leukemia-associated sequence variations with a VAF of at least 2%, and without associated cytopenia, dysplasia, or neoplasia.[8] This VAF corresponds to >4% of circulating blood cells being derived from a single clone, assuming heterozygosity of the driver event and a neutral copy number state. Age-related CH is a related term used to emphasize its strong link to the aging process.[23] The presence of unexplained and persistent cytopenias among individuals with CHIP sequence variations is termed clonal cytopenia of undetermined significance. Both CHIP and clonal cytopenia of undetermined significance are recognized as premalignant conditions by the World Health Organization characterization of myeloid neoplasms. The risk of progression to blood cancer is variable and may be estimated by the CH risk score.[24,25]

Herein, we use the term CH rather than CHIP to reflect a broader framework that encompasses all relevant consequences and implications. Most of our current knowledge of CH is limited to somatic sequence variations with a VAF of ≥2%, which approaches the error rate of standard whole exome sequencing methodologies.[26] The increasing availability of error-corrected deep sequencing methods enables improved accuracy of CH genotyping and detection of CH with lower VAF,[21] which may influence future VAF thresholds to define CH. Application of these technologies has revealed that CH driver sequence variations often arise early in life, with various exposures influencing the fitness of genetic variants across the life course determining the emergence of CHIP.[11] Longitudinal studies using such technologies are beginning to improve our understanding of the factors that influence CH initiation and expansion over the lifetime of an individual.[27,28]

RISK FACTORS FOR CH

Although age is the dominant risk factor for CH, several factors contributing to the risk of developing CH have emerged in the past decade (Supplemental Table 1).[6,27,29–32] These factors increase the risk for CH either by increasing DNA mutagenesis and thereby increasing the likelihood of acquiring CH-defining sequence variations or by conferring a relative fitness advantage to variant clones facilitating their growth. Herein, we focus on 3 broad categories of CH risk factors beyond chronologic age: germline genetic predisposition, toxic exposures such as chemotherapy, and chronic inflammation. Understanding these risk factors is crucial for both identifying high-risk individuals and developing strategies to mitigate CH-associated cardiovascular risk. Targeted interventions aimed at these risk factors may offer new avenues for preventing and managing CH-related CVD.

Germline Predisposition

Recent studies have highlighted how inherited common and rare genetic variants shape the landscape of CH.[33] Recent genome-wide association studies have collectively identified numerous common genetic variants that predispose individuals to CH.[29,30] The mechanistic link between these germline variants and CH development likely involves regulation of HSC function, DNA repair, and genomic stability. The most prominent is a common variant in the *TERT* locus, which increases CH risk with an odds ratio of 1.3.[34] *TERT* encodes telomerase reverse transcriptase, a key enzyme in maintaining telomere length.[35] Other loci associated with either increased or reduced CH risk include variants in *ATM* (odds ratio, 1.46 [1.33–1.60]) or *PARP1* (odds ratio, 0.87 [0.84–0.90]).[30,36] *ATM* is a key node in the DNA damage response (DDR) pathway; *PARP1* plays a crucial role in DNA repair.

Rare pathogenic variants, like those that give rise to inherited cancer genetic syndromes, can also predispose to CH. Recent work has linked rare pathogenic variants in *RUNX1*, *CHEK2*, *ATM*, *TINF2*, and *PTPN11*—a convergence of common and rare variants targeting the same genes—to a 2- to 10-fold increase in CH risk. Down syndrome, a congenital disorder caused by trisomy 21, also increases the risk of early-onset CH, which may reflect features of premature aging inherent to the syndrome.[31]

Some germline variants show gene-specific effects. For example, a common variant in the *TCL1A* promoter protects against *TET2* CH and *ASXL1* CH, but predisposes carriers to *DNMT3A* CH.[29,30] Recent experimental work demonstrated that sequence variations in *TET2*, *ASXL1*, and other epigenetic regulator genes implicated in CH, but not *DNMT3A,* increase chromatin accessibility at the *TCL1A* locus.[37] This leads to increased expression of *TCL1A*, a proto-oncogene implicated in leukemia, promoting HSC proliferation. A germline variant in the *TCL1A* promoter restricts this expression, thereby limiting CH expansions. Because *DNMT3A* is a relatively weak driver of CH expansion, germline *TCL1A* promoter variants that prevent the outgrowth of other CH clones lead to relatively more *DNMT3A* CH.[38] Patients with inherited bone marrow failure disorders are also at an increased risk for CH, which may arise in these circumstances either because of inherited defects in DNA damage repair or in some cases because the acquired CH sequence variation normalizes the germline deficit and relieves baseline HSPC fitness constraints.[39] Recent data suggest that ancestry may influence CH susceptibility. In a Mexico City cohort, CH frequency was 1.6-fold lower than that observed in the UK Biobank, with an even more pronounced reduction in *JAK2*-variant CH.

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Whether these disparities are attributable to underlying germline genetic variation, environmental exposures, or other correlated nongenetic factors remains to be determined.[40] These and other associations highlight the complex interplay between germline genetics and somatic sequence variations in CH development.

Toxin Exposure

Patients with cancer, particularly those exposed to cytotoxic therapies, have an increased prevalence of CH.[19,41] This phenomenon, termed therapy-related CH (t-CH), is frequently characterized by sequence variations in the DDR pathway genes, such as *TP53*, *PPM1D*, *CHEK2*, and *ATM*.[42] Chemotherapy acts as a strong selective pressure, conferring a fitness advantage to HSCs harboring sequence variations in the DDR pathway genes, therein enhancing survival under genotoxic stress.[19,43] The prevalence of t-CH can be as high as 30% even in relatively young cancer survivors, significantly higher than in age-matched controls.[44] CH is an emerging and important risk factor in the practice of cardio-oncology, particularly for long-term cancer survivors.

Radiation as a component of cancer treatment leads to 2- to 4-fold increased risk of CH, with risk being dependent on both radiation modality and intensity.[45] Studies have documented increased risk among individuals with higher exposures to radon.[46] How space radiation affects CH prevalence is an area of active interest in the community, with implications for cardiovascular health of astronauts embarking on future exploration missions.[47]

Smoking is another well-established risk factor for CH.[48] Smokers show an increased prevalence of *ASXL1*-variant CH.[49] Multiple mechanisms, including both direct DNA damage and chronic inflammation, are likely. CH was significantly more common in World Trade Center first responders, implicating particulate matter as a potential driver.[50] However, broader associations between air pollution and CH risk have not been established, warranting further investigation.[51]

Chronic Inflammation

Numerous studies in model organisms have demonstrated links between bone marrow inflammation and HSC health.[52,53] Chronic inflammation, particularly involving cytokines such as those in the IL-6 (interleukin-6) axis, can promote the expansion of variant HSCs and confer their clonal fitness.[54,55] In turn, dominant clones may activate anti-inflammatory programs to survive the inflammatory signals produced by their own mature cells, creating a self-perpetuating cycle that supports their continued growth. Multiple sources of inflammation have been linked to CH in humans. Human immunodeficiency virus infection is associated with an increased risk of CH.[56] The chronic immune activation and inflammation characteristic of human immunodeficiency virus infection, such as activated IFN-γ (interferon-γ) signaling, is hypothesized to create a permissive environment for the expansion of variant HSC clones.[57] Women with premature menopause have an increased risk of CH.[58] Loss of the protective effects of estrogen on the bone marrow microenvironment may contribute to accelerated aging and increased inflammation, mediating this increased risk. Multiple lifestyle factors related to chronic inflammation have been linked to CH, including obesity and sleep deprivation.[59,60] These conditions are associated with chronic low-grade inflammation and oxidative stress, which may promote DNA damage and provide a selective advantage to variant clones. A healthy diet was found to be protective against CH risk.[61] However, the association between chronic inflammation and CH risk appears to be context-dependent. Sickle cell disease—despite driving chronic inflammation and being a known myeloid malignancy risk factor—has not been linked to increased CH risk.[62]

CH AND CVD

The association between CH and CVD has been extensively studied in atherosclerotic CVD (ASCVD), with numerous studies confirming this relationship.[7] Studies using diverse case–control cohorts and large population-based cohorts have shown that the strength of the association varies depending on the specific variant genes, the types of sequence variations, clone size or VAF, and baseline CVD risk. Older CH carriers with sequence variations in *DNMT3A*, *TET2*, or *ASXL1* had a 1.7- to 2-fold increased ASCVD risk compared with those without CH.[63] Coronary artery calcium scoring with cardiac computed tomography, a well-validated noninvasive measure of coronary atherosclerosis, showed significantly higher CAC scores in CH carriers compared with noncarriers.[63] Among individuals who had early-onset myocardial infarction, CH was enriched 4-fold compared with older controls.[63] Moreover, patients with CH were found to have worse outcomes after myocardial infarction and cardiogenic shock compared with those without CH.[64] In the low-risk UK Biobank population, the association between CH and ASCVD has been inconsistent, in part due to variability of CH definition and calling as well as potential host-related factors and heterogeneity in sequence variation effects.[30,36,65] Other data sets have demonstrated more consistent associations between CH and ASCVD,[66] highlighting greater ASCVD risks particularly attributed to *TET2* and *JAK2* sequence variations or larger clones with VAF ≥10%. Nevertheless, the link between CH and recurrent ASCVD events among people with established ASCVD remains inconsistent.[66,67]

More recently, the directional relationship between CH and ASCVD has been explored. Computational modeling has suggested that CH may be a consequence of

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ASCVD itself, given the established literature on hematopoiesis in response to atherosclerosis.[68] However, longitudinal studies combining serial assessments of CH with imaging of atherosclerosis and clinical evaluations of ASCVD have demonstrated that atherosclerosis does not appear to drive the development of CH. As such, CH may be unidirectionally associated with the progression of atherosclerosis.[27,28]

Emerging evidence also points to associations between CH and heart failure (HF), including the differential impact of specific CH genes. In a meta-analysis of 56 597 individuals without HF at baseline, CH carriers had a 25% higher risk of developing HF, independent of traditional risk factors, and also among those without coronary artery disease. CH overall was not significantly associated with reduced ejection fraction, but gene-specific analysis showed that sequence variations in *ASXL1* were associated with modestly reduced ejection fraction.[69] A separate study involving 8090 participants from 2 prospective cohort studies found a higher cumulative incidence of HF among CH carriers compared with noncarriers.[70] CH overall was not significantly associated with either HF with preserved or reduced ejection fraction, but gene-specific analysis demonstrated that *TET2* sequence variations were 2.4-fold enriched among individuals with HF with preserved ejection fraction versus without. In another study that included patients with HF, CH was linked to a significantly higher risk of all-cause mortality, HF-specific death, and HF-related hospitalizations.[71] Additional studies have also reported that patients with CH and chronic HF had worse survival and prognosis than those with HF but without CH.[72]

Beyond ASCVD and HF, growing evidence suggests a broader association between CH (with variation by gene) and other cardiovascular disorders, including atrial fibrillation,[73] stroke,[74] peripheral artery disease,[75] aortopathy,[76] and myocarditis.[76a] In addition, recent data suggest that the presence of CH portends a worse prognosis in patients with aortic stenosis after valvular interventions.[77] CH, particularly involving *JAK2* sequence variations, has been linked to an increased risk of thrombosis.[78] A detailed summary of cardiovascular outcomes associated with CH can be found in Supplemental Table 2.

PATHOGENESIS AND PROPOSED MECHANISMS

The known genetic drivers of CH involve genes that are generally recurrently altered in hematologic malignancies. The most prevalent CH sequence variations occur in *DNMT3A*, *TET2*, and *ASXL1*, which encode epigenetic regulators that can widely affect chromatin structure and transcription factor accessibility. Sequence variations in these genes often result in loss of function, altering HSPC proliferation, self-renewal, or survival. Emerging evidence also suggests that specific sequence variation sites—particularly missense sequence variations—within CH-driver genes may confer differential functional consequences.[79] Mechanistic investigations into the association between CH and CVD have mainly used mouse models transplanted with bone marrow cells carrying surrogates of human CH sequence variations (Supplemental Table 3). Several variant genes have been shown to contribute to CVD by primarily exacerbating inflammatory responses, which leads to accelerated atherosclerosis, profibrotic cardiac remodeling, and cardiomyocyte dysfunction.[80] Thus far, the relative importance of systemic versus direct effects of variant immune cells in the myocardium or atherosclerotic plaques remains unclear.[81] In mouse models, Y chromosome–deficient cardiac macrophages have been shown to activate profibrotic signaling cascades, resulting in fibroblast activation, extracellular matrix deposition, and impaired myocardial function, thereby supporting a direct pathogenic role within the myocardium.[82] Further studies are needed to delineate the relative contributions of systemic versus local mechanisms in humans. In addition, although there are some shared inflammatory mechanisms connecting CH to CVD, several distinct mechanisms have been identified for the different driver genes, establishing the basis for potential personalized preventive strategies tailored to carriers of specific CH genotypes.

The effects of *TET2* sequence variations on CVD are the best characterized to date. Mouse models of *Tet2*-variant CH exhibit accelerated atherosclerosis and cardiac dysfunction in different settings, driven by heightened production of proinflammatory cytokines and chemokines by variant macrophages, predominantly, but not exclusively, IL-1β (interleukin 1β) and related cytokines.[63,83,84] *Tet2* inactivation significantly increases *Il1β* transcript levels in macrophages exposed to various proinflammatory stimuli,[83–85] partially mediated by increased histone acetylation at the *Il1β* promoter.[83] In addition, TET2 (Tet methylcytosine dioxygenase 2)–deficient macrophages exhibit an overactivation of the NLRP3 (Nod-like receptor protein 3) inflammasome, the main regulator of IL-1β maturation and secretion in CVD,[83–87] leading to an increase in IL-1β secretion.[83,85,86] In correspondence, humans carrying somatic *TET2* sequence variations, but not other CH sequence variations, exhibit elevated circulating levels of IL-1β.[29] These findings strongly suggest that targeting IL-1β–driven inflammation may be particularly effective for preventing CVD in *TET2* sequence variation carriers. Supporting this possibility, pharmacologic inhibition of NLRP3 or the related cytokine IL-6 has been shown to suppress the effects of *Tet2*-variant cells on various mouse models of CVD.[83,84,87] Consistent observations have been made in humans, where an inherited genetic variant that reduces IL-6 signaling (*IL6R* D358A) mitigates the elevated coronary artery disease risk of *TET2* sequence variation

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carriers.[65] Moreover, a post hoc analysis of CANTOS (Canakinumab Anti-Inflammatory Outcomes Study) suggests that canakinumab (an IL-1β–neutralizing antibody) versus placebo reduces the risk of ischemic events to a much greater extent in carriers of *TET2* sequence variations than in those without CH.[88] Colchicine, another anti-inflammatory drug known to inhibit IL-1β–driven inflammation, has been shown to mitigate *TET2*-variant CH–driven atherosclerosis.[89]

Dnmt3a loss-of-function sequence variations have also been shown to promote atherosclerosis and cardiac dysfunction by exerting similar, albeit milder, effects compared with *Tet2* sequence variations on the expression of several proinflammatory cytokines and chemokines in murine macrophages.[90] Furthermore, mouse models of *Dnmt3a*-variant and *Tet2*-variant CH harbor distinct tissue resident-like macrophage populations and inflammatory gene expression signatures.[90] The specific molecular regulatory pathways underlying the proinflammatory effects of *Dnmt3a* sequence variations remain to be determined. In addition, hematopoietic *Dnmt3a* loss of function impairs efferocytosis within atherosclerosis plaques, which may contribute to adverse plaque remodeling,[91] and increases secretion of HB-EGF (heparin-binding epidermal growth factor–like growth factor), which induces cardiac fibrosis.[92] Recent studies also highlight the potential involvement of diverse variant immune cells beyond monocytes and macrophages, including CD4+ T cells and natural killer cells, as well as the role of paracrine signaling pathways between variant and wild-type cells, in driving and amplifying inflammatory cascades observed in CH-mediated HF.[93]

Unlike the inactivating sequence variations that occur in *TET2* and *DNMT3A*, *ASXL1* sequence variations predominantly occur in the C-terminal exon that encodes a regulatory domain, leading to the expression of a truncated protein. *ASXL1* is substantially less studied in the context of CH and CVD, but *Asxl1*-variant CH promotes inflammation and cardiac remodeling after left anterior descending ligation in mice, like *Tet2*- and *Dnmt3a*-variant CH,[94] and has been found to accelerate atherosclerosis through enhanced myeloid cell inflammation.[95]

The *JAK2* V617F variant, a common driver of myeloproliferative neoplasms, has been strongly associated with an increased risk of CVD, myocardial infarction, and arterial and venous thromboembolic risk.[6,96,97] In many studies of CH, *JAK2* V617F variant CH is associated with increased platelet and red blood cell counts, suggestive of early myeloproliferative neoplasms rather than benign clonal expansion, but identification of *JAK2* V617F in 3% to 4% of a general European population by droplet digital polymerase chain reaction–based targeted sequencing (with detection sensitivity of 0.0009%)[98] suggests the variant may be associated with increased CVD risk even in the absence of increased blood cell counts.[97] In mouse models of CH, *Jak2* V617F accelerates venous and arterial thrombosis,[96] HF after left anterior descending ligation,[99] atherosclerosis,[100] and thoracic aortic aneurysm,[101] through a variety of mechanisms. Thrombotic risk is driven by increased platelet activation, polycythemia, and neutrophil extracellular trap formation,[96,97] whereas HF involves increased myeloid cell and myocardial inflammation.[99] Mechanisms of accelerated atherosclerosis of *Jak2* V617F CH include increased macrophage proliferation, mitochondrial reactive oxygen species formation, augmented DNA damage, NLRP3 and AIM2 (absent in melanoma 2) inflammasome activation, increased erythrophagocytosis and ferroptosis, and impaired efferocytosis.[97,100]

t-CH involving sequence variations in *TP53*, *PPM1D*, *ATM*, or *CHEK2* may occur after cytotoxic therapy for cancer. The expression of these genes can be induced by genotoxic stressors, which are believed to promote the selective expansion of small preexisting clones that harbor DDR pathway sequence variations. Individuals with t-CH are at increased risk of developing therapy-related myeloid neoplasms, and experimental studies suggest that DDR-variant CH may contribute to the late cardiotoxicity observed in patients who have been treated for cancer.[102] One such study examined the role of *Tp53*-mediated CH in a mouse model of anthracycline-induced cardiomyopathy[103] and found that doxorubicin promoted the expansion of *Tp53*-variant hematopoietic cells, leading to worse cardiotoxicity. Another study demonstrated that inactivating sequence variations in *Tp53* accelerated atherosclerosis and increased plaque macrophage burden, mainly due to accelerated macrophage cell cycle kinetics and proliferation.[75] Although not directly evaluated in the context of atherosclerosis or other vascular conditions, *Ppm1d* sequence variations linked to CH have been shown to promote fibrotic remodeling of the myocardium in mice infused with angiotensin II.[104] Similar to other CH driver genes, *Ppm1d* sequence variations are associated with increased expression of IL-1β, and the pathologic effects of variant *Ppm1d* on the heart can be reversed by treatment with an NLRP3 inflammasome inhibitor in experimental models.[104]

CH can also result from mosaic chromosomal alterations of autosomes and sex chromosomes, although its association with CVD remains unclear, with the exception of loss of Y chromosome.[82] Mechanistic studies are challenged by the size of the lesions and the individual-to-individual heterogeneity of these molecular events. CH sequence variations in genes such as RNA-splicing factors (*SF3B1*, *SRSF2*, *U2AF1*, and *ZRSR2*) are rare and have yet to be explored in experimental models of CVD. Furthermore, studies have reported the existence of frequent hematopoietic clonal events that cannot be attributed to easily identifiable sequence variations in known driver genes, known as CH with unknown drivers.[105] It is conceivable that some of these clonal events do not have a variant-driven cause,[34] which makes it challenging

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to develop experimental models or conduct mechanistic investigations.

CLINICAL TRANSLATION

As our understanding of CH continues to evolve, how these insights may be integrated into clinical practice must be considered. Successful clinical translation would require targeted screening strategies to identify at-risk patients with CH, tailored risk prediction models, effective surveillance and therapeutic strategies, and collaborative, multidisciplinary teams to address the diverse health effects of CH.

No official guidelines exist for screening for CH sequence variations. CH is identified incidentally when DNA sequencing is used for other purposes, such as hereditary cancer panel testing, solid tumor evaluations involving circulating tumor or cell-free DNA, and when hematology sequencing panels are used in patients with abnormal blood counts suspected of having a hematologic malignancy.[106] Several research groups have begun prospective screening studies aimed at evaluating the clinical consequences of CH.[107] More research is warranted to devise targeted sequencing strategies, particularly for at-risk groups (eg, cancer survivors), and to define appropriate screening intervals, which could enable earlier detection and intervention.

CH risk stratification tools have emerged that assist clinicians in estimating the likelihood of evolution to myeloid neoplasm in patients with CH.[24] One such tool, the CH risk score, stratifies patients as being at high, intermediate, or low risk for myeloid neoplasm progression, based on factors such as the number of CH sequence variations, clonal size (VAF ≥0.2), and the presence of high-risk sequence variations, among others. Individuals at high risk have a >50% chance of developing blood cancer in 10 years and are monitored more closely than individuals at low risk, who have a <1% risk of the same outcome.[24] Subsequent analyses using the CH risk score has demonstrated that the model may also be useful for predicting risk of CVD-related death in older adults.[108] However, the relationship between myeloid malignancy risk and CVD risk in the context of CH remains incompletely understood, supporting the need for a dedicated risk model for CH-associated CVD accounting for the competing risk of myeloid malignancy. Given the dynamic nature of CH, assessing clonal evolution through serial sampling may offer additional prognostic values.

Several tertiary medical centers have developed CHIP clinics for multidisciplinary management and counseling of patients with CH.[109] These clinics develop risk-specific surveillance and management strategies, primarily focusing on hematologic malignancy, and CH-associated cardiovascular risk may also be evaluated by cardiology or cardio-oncology. Current population-based cardiovascular risk prediction models do not incorporate CH, and no targeted therapies exist to mitigate CH-associated cardiovascular risk. Therefore, individualized risk assessment and optimization of the cardiovascular risk profile using guideline-concordant primary and secondary prevention are the preferred management strategies for patients with CH.[110] Beyond cancer and CVD risk, the broad health implications of CH, such as chronic lung disease, fatty liver disease, and endocrinopathies, underscore the necessity for a multidisciplinary approach. As our knowledge of CH advances, these clinics may play an essential role in translating research discoveries into practical, patient-focused care strategies that address wide-ranging health implications of CH.

Clinical trials and interventions in CH are beginning to emerge (Figure 2). Studies to date have focused on myeloid malignancy prevention and targeting high-risk CH; future clinical trials evaluating the efficacy of interventions in CH to mitigate CVD risk are an attractive next step. Therapies targeting specific hematopoietic clones by restoring CH gene function or attenuating clonal expansion may offer potential benefit; vitamin C may restore TET2 function and metformin may reduce *DNMT3A*-variant clonal fitness.[111–113] Given the inflammatory sequelae of CH, anti-inflammatory strategies such as inflammasome or cytokine inhibition may prevent cardiovascular risk in patients with CH. Considering the modest but significant infection risk observed in CANTOS, maximizing cardiovascular benefit while ensuring infection surveillance may confer a net therapeutic advantage.[88] When designing trials targeting CH-mediated malignancy or CVD risk, incorporating cardiovascular phenotyping into cancer studies, or malignancy surveillance into cardiovascular studies, may be beneficial, given their shared risk factors and mechanisms.

CONCLUSION AND FUTURE OUTLOOK

CH is increasingly recognized as a potential driver of CVD. Whereas substantial progress has been made in establishing the association of CH with various CVDs, the role of CH as a causal factor, associative marker, or both remains under investigation. In addition, the precise gene-specific mechanisms driving these outcomes remain incompletely understood. Further research should aim to better define the molecular mechanisms through which these sequence variations promote cardiovascular pathology and contribute to adverse cardiovascular outcomes. Advances in genomic technologies, such as single-cell transcriptomic profiling of peripheral blood cells, may help uncover the functional consequences of CH driver sequence variations and inform targeted therapeutic strategies. In addition, because clonal size has emerged as a key determinant of CH-related CVD, with larger clones linked to greater risk, longitudinal studies with serial blood sampling can offer a more comprehensive understanding of CH dynamics over time, allowing

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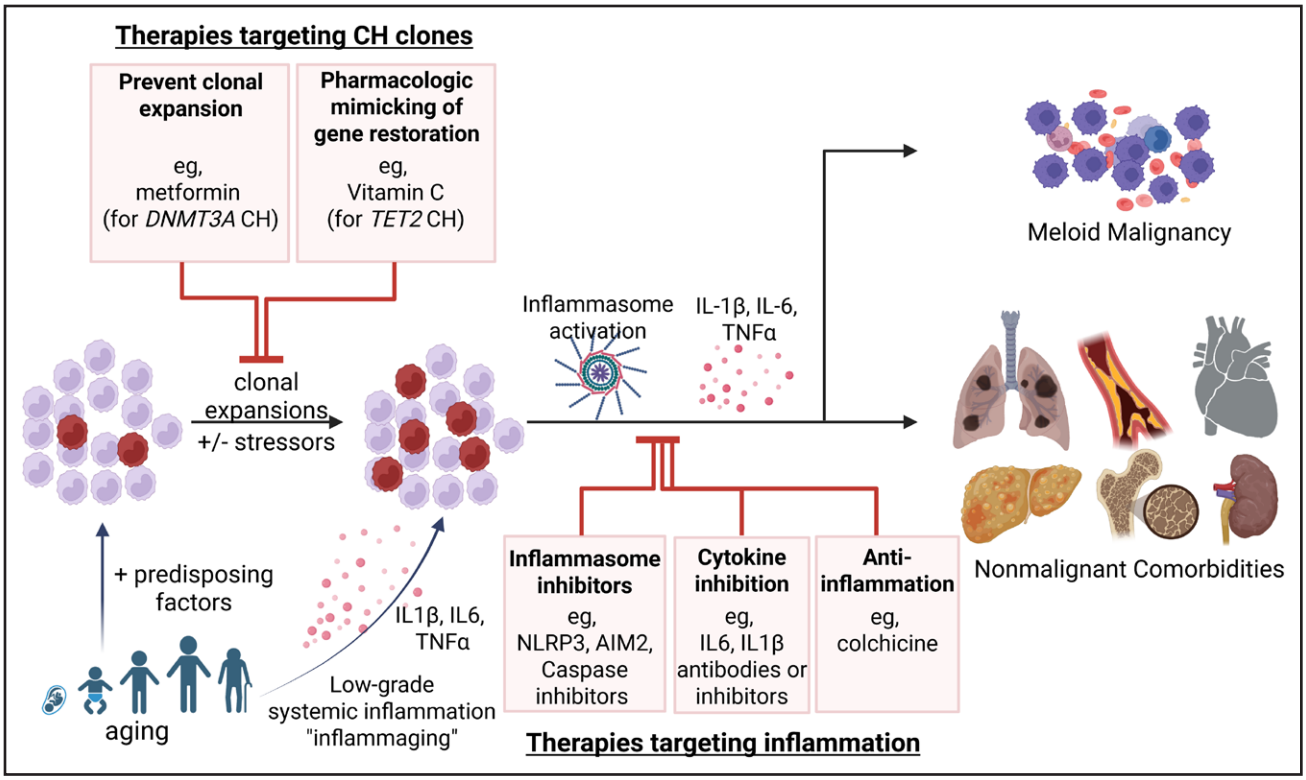

Figure 2. Potential treatment strategies for clonal hematopoiesis and associated risks.
Clonal hematopoiesis (CH) is associated with aberrant expression of proinflammatory cytokines, which are dually involved in the pathogenesis of myeloid malignancies and nonmalignant comorbidities, such as cardiovascular disease. Therapeutic prevention strategies are in the early stages of clinical development, with targets including CH clonal reduction and anti-inflammatory targets such as inflammasome pathways and specific cytokines. Figure created with BioRender.com.

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for interrogation of the relationship between changes in clone size and CVD risk. Therapeutic approaches focusing on reducing inflammation hold promise, but prospective clinical studies are necessary to confirm their effectiveness in lowering CH-associated CVD risk. Future research should also explore novel strategies aimed at slowing clonal expansion, thereby preventing disease progression. Incorporating the assessment of CH status into clinical practice is another key step. Development of CVD risk prediction models that integrate CH with traditional cardiovascular risk factors may help clinicians more accurately identify patients at high risk for CVD and tailor therapies accordingly. This personalized approach could allow for earlier and more targeted interventions that reduce the burden of CVD.

ARTICLE INFORMATION

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Disclosures

Writing Group Disclosures

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This table represents the relationships of writing group members that may be perceived as actual or reasonably perceived conflicts of interest as reported on the Disclosure Questionnaire, which all reviewers are required to complete and submit. A relationship is considered to be "significant" if (a) the person receives ≥$5000 during any 12-month period, or ≥5% of the person's gross income; or (b) the person owns ≥5% of the voting stock or share of the entity, or owns ≥$5000 of the fair market value of the entity. A relationship is considered to be "modest" if it is less than "significant" under the preceding definition.

*Modest.

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Reviewer	Employment	Research grant	Other research support	Speakers' bureau/ honoraria	Expert witness	Ownership interest	Consultant/ advisory board	Other
Aarti Asnani	Beth Israel Deaconess Medical Center & Harvard Medical School	NIH (PI on NIH R01 and co-I on 4 NIH R01s)†; Genentech (PI on sponsored research agreement)†	None	UpToDate*	None	Corventum, Inc: equity and patent royalties†	None	None
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*Modest.

†Significant.

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